

SHAURYA PUROHIT

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PROFESSIONAL SUMMARY

I am proficient and have good hands-on experience in the fields of Artificial Intelligence, Machine learning, Deep Learning, Federated Learning, Cybersecurity, Data Analysis, and Full Stack Development. My research is centered on creating cutting-edge AI and ML solutions for Anomaly and Intrusion Detection Systems, specifically tailored to enhance the security of critical infrastructure through state-of-the-art methodologies. I have a strong foundation in data privacy, statistical modelling for large datasets, data preprocessing, feature engineering, and model evaluation, with practical experience in Python, TensorFlow, Keras, and other ML frameworks.

EDUCATION

Pursuing **PhD in Computer Engineering**
Iowa State University - Ames, Iowa
CGPA: 3.86/4.00

August 2019 – February 2025
(Expected)

Master of Engineering in Computer Engineering (Minor in Statistics)
Iowa State University - Ames, Iowa
CGPA: 3.84/4.00

August 2019 - May 2022

Bachelor of Technology in Information Technology
Manipal University Jaipur - Jaipur, India
CGPA: 9.58/10.00

August 2015 - May 2019

PROFESSIONAL EXPERIENCE

Iowa State University - Ames, Iowa

Graduate Research Assistant

May 2020 – Present

- Starting Spring 2024, started working on two projects in collaboration with Argonne National Laboratory
 - The first project enhances real-time operations of distributed energy resources, providing cost-effective, real-time cybersecurity for individual DERs within an aggregation pool through Reinforcement learning and Game theory.
 - The second aims to bolster grid cybersecurity through a novel Machine and Federated Learning methodology. The project underscores a commitment to integrating Machine Learning to address complex energy and security challenges through threat intelligence and anomaly detection.
- Worked on Cybersecurity of DER communication protocols and EV Charging using Federated Learning.
 - Developed federated learning anomaly detection models enhancing data privacy and cybersecurity in DER and EV charging networks, achieving high accuracy and reduced loss on publicly available large datasets.
- Had the honor of being **selected as a senior PhD student to serve on a panel at the CAE-R Symposium 2023**, where I presented both my research and poster.
- Worked on Game-theoretic Optimization of Cybersecurity Investment Strategies for EV Charging Stations using Attack Defense Trees.
 - This framework integrates the CIA triad and MITRE ATT&CK framework to analyze attack paths, optimize resource allocation, and formulate robust defense strategies.
- Worked on cybersecurity of autonomous vehicles using Machine Learning & Deep Neural Networks
 - Developed a Hybrid Anomaly detection system for preventing cyber-attacks on intra-vehicular communication networks with the help of rule based and machine learning techniques. Achieved an accuracy of around 95%.
 - Submitted the enhanced journal version of this work using Deep Neural Networks and variety of datasets. Achieved an accuracy of more than 97% and better latency.
- Worked on a project titled Cloud-based multi-sensor remote data acquisition system.

Mutual of Omaha – Nebraska, USA
Information Technology Intern

May 2022 – December 2022

- Worked as an I/S intern with Security and Data Communication team.
 - Worked on Deep Neural Networks for Forcepoint Framework.
 - Worked on complete automation and full stack management of an in-house project titled “Moonet” using Jenkins CI/CD pipeline architecture.

Iowa State University - Ames, Iowa
Graduate Teaching Assistant

August 2021 - December 2021

- Worked as a TA for a 500-level course: Real-Time systems (**CPR E 558**)
- Responsibilities included: one-to-one interaction with students for solving queries, helping students in their final course project, providing feedback by grading assignments, quizzes, and exams.

Graduate Teaching Assistant

August 2019 - May 2020

- Worked as a TA for courses: Introduction to Computer Engineering and Problem Solving (**CPR E 185**) & Problem Solving in Software Engineering (**S E 185**).
- Responsible for conducting labs every week to help students write programs in C, helping students in their final course project, holding office hours to facilitate one-to-one interaction for resolving queries, grading homework, lab reports, exams and providing appropriate feedback.

ValueLabs – Hyderabad, India

January 2019 - April 2019

Software Engineer Intern

- Worked as a Full-Stack Developer on a real-time client project titled MfExpert (Micro- Finance Expert) and designed a web application using Angular 7, Node.js and PostgreSQL.

IIT BHU – Varanasi, India

May 2018 - July 2018

Machine Learning Intern

- Worked on a project titled Autonomous Car Detection using Deep Learning under the guidance of Dr. Tanima Dutta.
- Implemented Car Detection in images and videos using Machine Learning and Deep Learning and achieved a high accuracy of 98%.

CERTIFICATIONS

- Machine Learning
Issued by: Stanford University (via Coursera)
- Cybersecurity Specialization
Issued by: IBM (via Coursera)
- Cybersecurity Fundamentals
Issued by: IBM
- AWS Machine Learning
Issued by: Amazon Web Services (via Coursera)
- Programming in Java
Issued by: NIIT Technologies

PUBLICATIONS

Conferences:

- **S. Purohit** and M. Govindarasu, "ML-based Anomaly Detection for Intra-Vehicular CAN-bus Networks," 2022 IEEE International Conference on Cyber Security and Resilience (CSR), Rhodes, Greece, 2022, pp. 233-238, doi: 10.1109/CSR54599.2022.9850292. (**Published**)
- **S. Purohit** and M. Govindarasu, "Cybersecurity Investment Analysis for Electric Vehicle Charging Infrastructures," 2023 Resilience Week (RWS), National Harbor, MD, USA, 2023, pp. 1-6, doi: 10.1109/RWS58133.2023.10284670. (**Published**)
- **S. Purohit** and M. Govindarasu, "FL-EVCS: Federated Learning based Anomaly Detection for EV Charging Ecosystem," 2024 33rd International Conference on Computer Communications and Networks (ICCCN), Kailua-Kona, HI, USA, 2024, pp. 1-9, doi: 10.1109/ICCCN61486.2024.10637543. (**Published**)
- **S. Purohit** and M. Govindarasu, "Federated Learning based Anomaly Detection for Distributed Energy Resource Communication," 2024 IEEE Power & Energy Society General Meeting (PESGM), Seattle, WA, USA, 2024, pp. 1-5, doi: 10.1109/PESGM51994.2024.10688634. (**Published**)
- B. Blakely, **S. Purohit**, K. Ogbuefi, I. Prieto, and M. Govindarasu. "A Federated Intrusion Response Network for Grids (FIRNet-G)." IECON 2024. (**Accepted**)
- K. Wang, **S. Purohit** and M. Govindarasu, "A Robust Framework for Cyber Attack Detection and Mitigation in DER Networks using Reinforcement Learning and Game Theory, " 2025 IEEE Power & Energy Society General Meeting (PESGM) (**Accepted**)
- P. K. Mallaiah, N. Saqib, S. Purohit and M. Govindarasu, "ML-based Multiclass Anomaly Detection System for Charge Profile Manipulation Attacks in EV Charging Ecosystem, " 2025 IEEE Power & Energy Society General Meeting (PESGM) (**Submitted- Under Revision**)

Journals:

- **S. Purohit**, M. Govindarasu and B. Blakely, "FL-ADS: Federated Learning Anomaly Detection System for Distributed Energy Resource Networks", " IET Cyber-Phys. Syst., Theory Appl. e70001(2025), doi: 10.1049/cps2.7000118. (**Published**)
- **S. Purohit** and M. Govindarasu, "HAVEN: A Hybrid Anomaly Detection System for Intra-Vehicular CAN-bus Communication using Rule-based and Neural Networks" , IEEE Transactions on Dependable and Secure Computing. (**Submitted- Under Revision**)

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, R
- **Web Based Technologies:** HTML, CSS, Angular, Node.js, PHP, JavaScript, jQuery, Ajax, XML, DHTML
- **ML Frameworks:** Tensor flow, Keras, Scikit-learn, PyTorch, Flower
- **Cloud Platforms:** AWS, Elastic Cloud
- **Subject specific software:** Docker, RStudio, STAR UML, AutoCAD
- **Databases:** MySQL, PostgreSQL

ACHIEVEMENTS & AWARDS

- Secured **1st Rank in B. Tech (IT) 3rd year & 4th year** for the academic year 2017-18 and 2018-19 respectively.
- Secured **Statewide Ranking 1st & India Wide Ranking 3rd** in the Wheebox Employability Skills Test in 2017.
- Published article for e-magazine **EasyShiksha** in 2018.